# 1 Supplement

In this note we investigate properties of the function $V(w, f, T)$ defined by the following recurrence relation

$$V(w,f,T) = \max \begin{cases} \frac{\alpha}{\alpha+\beta} + \frac{\alpha}{\alpha+\beta} V(w+1, f, T-1) + \frac{\beta}{\alpha+\beta} V(w, f+1, T-1) \\ \frac{\alpha_0}{\alpha_0+\beta_0} + \frac{\alpha_0}{\alpha_0+\beta_0} V(1, 0, T-1) + \frac{\beta_0}{\alpha_0+\beta_0} V(0, 1, T-1) \end{cases}, \tag{1}$$

where $\alpha(w) = \alpha_0 + w$, $\beta(f) = \beta_0 + f$ and $V(w, f, 0) = 0$.

**Proposition 1.** *For any $w$ and $f$ we have $V(w+1, f, T) \geq V(w, f, T)$ and $V(w, f, T) \geq V(w, f+1, T)$.*

Let us consider the case $T = 1$. If one or both $V(w+1, f, 1)$ and $V(w, f, 1)$ equal $\alpha_0/(\alpha_0 + \beta_0)$ the inequality becomes trivial. The same can be said about about $V(w, f, T)$ and $V(w, f+1, T)$. Alternatively

$$\begin{aligned} V(w+1,f,T) - V(w,f,T) = \frac{\alpha(w)+1}{\alpha(w)+\beta(f)+1} - \frac{\alpha(w)}{\alpha(w)+\beta(f)} = \\ \frac{\beta(f)}{(\alpha(w)+\beta(f))(\alpha(w)+\beta(f)+1)} \geq 0, \end{aligned} \tag{2}$$

and

$$\begin{aligned} V(w,f,T) - V(w,f+1,T) = \frac{\alpha(w)}{\alpha(w)+\beta(f)} - \frac{\alpha(w)}{\alpha(w)+\beta(f)+1} = \\ \frac{\alpha(f)}{(\alpha(w)+\beta(f))(\alpha(w)+\beta(f)+1)} \geq 0. \end{aligned} \tag{3}$$

Let us now assume that $V(w+1, f, T) \geq V(w, f, T)$ and $V(w, f, T) \geq V(w, f+1, T)$. Then, again skipping trivial cases when one or both of the values $V(w+1, f, T+1)$ and $V(w, f, T+1)$ (or $V(w, f, T+1)$ and $V(w, f+1, T+1)$) are equal to $V(0, 0, T+1)$, we have

$$\begin{aligned} \frac{\alpha(w)+1}{\alpha(w)+\beta(f)+1} V(w+2, f, T) + \frac{\beta(f)}{\alpha(w)+\beta(f)+1} V(w+1, f+1, T) - \\ \frac{\alpha(w)}{\alpha(w)+\beta(f)} V(w+1, f, T) - \frac{\beta(f)}{\alpha(w)+\beta(f)} V(w, f+1, T) \geq \\ \frac{\beta(f)}{(\alpha(w)+\beta(f))(\alpha(w)+\beta(f)+1)} V(w+1, f.T) - \\ \frac{\beta(f)}{(\alpha(w)+\beta(f))(\alpha(w)+\beta(f)+1)} V(w+1, f+1.T) \geq 0. \end{aligned} \tag{4}$$

and

$$\begin{aligned} \frac{\alpha(w)}{\alpha(w)+\beta(f)+1} V(w+1, f, T) + \frac{\beta(f)}{\alpha(w)+\beta(f)} V(w, f+1, T) - \\ \frac{\alpha(w)}{\alpha(w)+\beta(f)+1} V(w+1, f+1, T) - \frac{\beta(f)+1}{\alpha(w)+\beta(f)+1} V(w, f+2, T) \geq \\ \frac{\alpha(w)}{(\alpha(w)+\beta(f))(\alpha(w)+\beta(f)+1)} V(w+1, f+1.T) - \\ \frac{\alpha(w)}{(\alpha(w)+\beta(f))(\alpha(w)+\beta(f)+1)} V(w, f+1.T) \geq 0. \end{aligned} \tag{5}$$

That immediately leads to $V(w+1, f, T+1) \geq V(w, f, T+1)$ and $V(w, f, T+1) \geq V(w, f+1, T+1)$. Then, by induction, $V(w+1, f, T) \geq V(w, f, T)$ and $V(w, f, T) \geq V(w, f+1, T)$ hold for any $T$.